Sentiment Analysis of Amazon Consumer Reviews

Dataset

The analysis uses Datafiniti_Amazon_Consumer_Reviews_of_Amazon_Products_May19.csv, a Datafiniti dataset of Amazon consumer product reviews. The supplied file contains 28,332 rows and 24 columns. The sentiment feature is reviews.text, the file also contains reviews.rating, which is used only as a proxy reference for optional evaluation. The rating distribution is 965 one-star, 616 two-star, 1,206 three-star, 5,648 four-star, and 19,897 five-star reviews.

1. Preprocessing and sentiment method

The script first validates the CSV path and confirms that reviews.text is present. It drops missing and blank review text, the supplied dataset had no missing values in this field, leaving 28,332 usable reviews. Each review is converted to a string, lowercased and stripped of surrounding whitespace. spacy tokenisation then removes punctuation, spaces, and stop words. Meaningful negators (for example, 'not' and 'never') are retained because removing them can reverse a review's meaning.

The program loads spacy's en_core_web_md model and adds the spacytextblob pipeline. For every cleaned review, it obtains blob.polarity. Scores above 0.05 are labelled Positive, scores below -0.05 Negative and all other scores Neutral. It also uses spacy's vector-based similarity() method to compare the first two sample reviews.

2. Testing and evaluation

The program prints three labelled sample predictions, including each source review, its cleaned text, the polarity score, and the predicted sentiment. This provides a direct sanity check that preprocessing and classification are working. The command-line option --evaluate (default 100 reviews) performs a reproducible proxy evaluation. It maps 1-2 stars to Negative, 3 stars to Neutral, and 4-5 stars to Positive, then prints the number of reviews evaluated, prediction accuracy, and predicted-class counts.

The completed run evaluated 100 reviews and achieved 71.0% proxy accuracy against the star-rating classes. It predicted 73 Positive, 18 Neutral and 9 Negative reviews. The first three sample results were Negative (-0.450), Negative (-0.500), and Positive (0.800). The first two reviews had a semantic similarity score of 0.732.

3. Strengths and limitations

Strengths include explicit input validation, modular functions, readable labelled output, preservation of negation, continuous polarity scores, semantic review comparison and a bounded evaluation sample that avoids unnecessary processing. The medium spacy model provides word vectors, making the similarity measure more meaningful than word overlap.

The polarity score is lexicon-based and can miss sarcasm, product-specific vocabulary, mixed opinions and context. Ratings are an imperfect ground truth because a star rating can express factors not stated in the review text, therefore the reported accuracy is a proxy metric, not a definitive model-performance measure. Cleaning can also remove useful context, although retaining negators reduces this risk. A stronger future approach would use manually labelled reviews or a held-out, domain-specific classifier.